

Triggers & Event Types

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Event-Driven Workflow Triggers

Triggers determine when workflows execute. This notebook covers all trigger types, Davis problem events, metric events, schedules, and custom event triggers.

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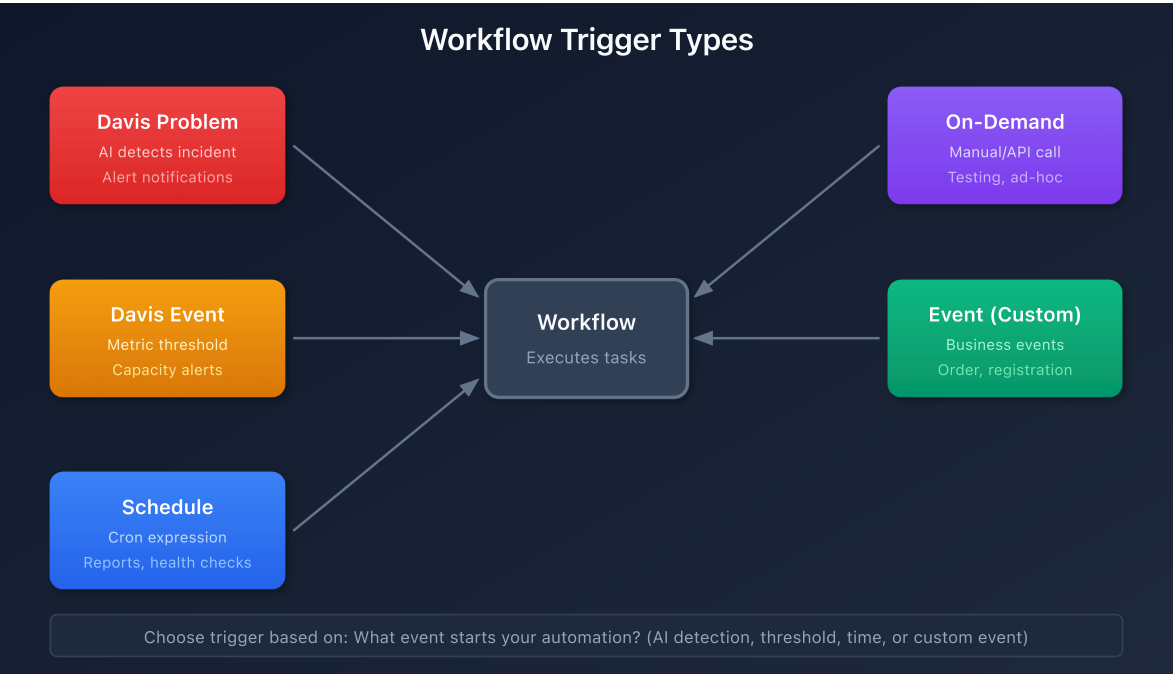
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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription
Permissions	<code>automation:workflows:write</code>
Prior Knowledge	WFLOW-01: Workflow Fundamentals

1. Trigger Types Overview

Trigger Type	Fires When	Primary Use Case
Davis Problem	Davis AI detects/updates/closes a problem	Alert notifications, incident management
Davis Event	Metric threshold breached	Capacity alerts, proactive notifications
Schedule	Cron expression matches	Reports, health checks, cleanup jobs
On-Demand	Manual execution or API call	Testing, ad-hoc automation
Event	Business/custom event ingested	Business process automation



Choosing the Right Trigger

Scenario	Recommended Trigger
"Notify when a service is slow"	Davis Problem
"Alert when CPU > 90% for 5 mins"	Davis Event
"Send weekly status report"	Schedule
"Process order completion events"	Event (bizevents)
"Test my workflow"	On-Demand

2. Davis Problem Trigger

The most common trigger for alert notifications. Fires when Davis AI detects problems.

Trigger Configuration

Setting	Description	Example
Categories	Problem types to include	Infrastructure, Application
Entity Type	Entity types to filter	Service, Host, Process
Management Zones	Scope to specific zones	Production, Checkout
Tags	Entity tags to match	<code>env:prod</code> , <code>team:checkout</code>

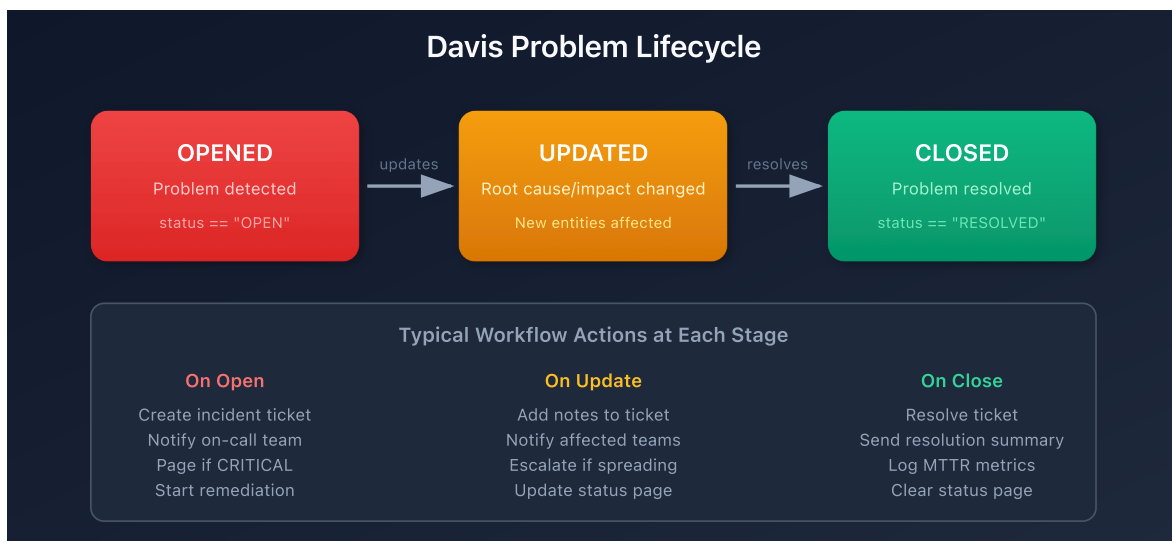
Problem Event Data

When a Davis problem triggers, you get access to:

```
{
  "display_id": "P-12345",
  "title": "High response time on checkout service",
  "severity": "CRITICAL",
  "status": "OPEN",
  "start_time": "2026-01-27T10:00:00Z",
  "affected_entity_ids": ["SERVICE-ABC123"],
  "root_cause_entity_id": "HOST-XYZ789",
  "management_zones": ["Production"],
  "impacted_entities": [...],
  "problem_url": "https://env.dynatrace.com/ui/problems/P-12345"
}
```

Problem Lifecycle Events

Event	When	Typical Action
Problem opened	New problem detected	Create incident, notify team
Problem updated	Root cause/impact changed	Update incident notes
Problem closed	Problem resolved	Close incident, send summary



Example: Filter Critical Production Problems

```
trigger:
  type: davis-problem
  config:
    categories:
      - AVAILABILITY
      - PERFORMANCE
    entityTagsMatch: all
    entityTags:
      - key: env
        value: prod
```

3. Davis Event Trigger (Metrics)

Trigger based on metric thresholds without creating a Davis problem.

When to Use

- **Proactive alerts** before Davis detects a problem
- **Capacity warnings** (disk 80%, memory usage)
- **Business KPIs** (orders/minute, cart abandonment)
- **Custom thresholds** different from Davis baselines

Configuration

Setting	Description
DQL Query	Metric query defining the threshold
Evaluation Frequency	How often to check (1m - 1h)
Alert Condition	When the threshold is breached

Example: CPU Warning at 80%

```
trigger:
  type: davis-event
  config:
    query: |
      timeseries avg_cpu = avg(builtin:host.cpu.usage), by:{dt.entity.host}
      | filter avg_cpu > 80
    evaluationFrequency: "5m"
```

Event Data

```
{
  "metric_key": "builtin:host.cpu.usage",
  "value": 85.5,
  "threshold": 80,
  "entity_id": "HOST-ABC123",
  "triggered_at": "2026-01-27T10:00:00Z"
}
```

4. Schedule Trigger

Execute workflows on a recurring schedule using cron expressions.

Cron Expression Format

```

|----- minute (0-59)
| |----- hour (0-23)
| | |----- day of month (1-31)
| | | |----- month (1-12)
| | | | |----- day of week (0-6, Sun=0)
| | | | |
* * * * *
```

Common Schedules

Schedule	Cron Expression	Description
Every hour	<code>0 * * * *</code>	Top of every hour
Daily at 9 AM	<code>0 9 * * *</code>	Every day at 9:00
Weekdays at 8 AM	<code>0 8 * * 1-5</code>	Mon-Fri at 8:00
First of month	<code>0 0 1 * *</code>	1st day, midnight
Every 15 minutes	<code>*/15 * * * *</code>	0, 15, 30, 45 past

Example: Daily Health Check Report

```
trigger:
  type: schedule
  config:
    cron: "0 9 * * 1-5" # Weekdays at 9 AM
    timezone: "America/New_York"
```

Schedule Best Practices

- **Avoid minute 0** - Many workflows run at :00, spread load
- **Use timezones** - Be explicit about timezone
- **Consider rate limits** - Don't schedule too frequently

5. On-Demand Trigger

Manual execution for testing, ad-hoc runs, or API-triggered automation.

Manual Execution

1. Open workflow in editor
2. Click **Run** button
3. Optionally provide input parameters
4. View execution results

API Execution

Trigger workflows programmatically via API:

```
curl -X POST
"https://<env>/platform/automation/v1/workflows/<id>/run" \
-H "Authorization: Api-Token <token>" \
-H "Content-Type: application/json" \
-d '{"params": {"custom_param": "value"}}'
```

Input Parameters

Define custom parameters for on-demand workflows:

```
trigger:
  type: on-demand
  config:
    parameters:
      - name: environment
        type: string
        required: true
      - name: notify_slack
        type: boolean
        default: true
```

Access in tasks:

```
{{ trigger().params.environment }}
{{ trigger().params.notify_slack }}
```

6. Event Trigger (Custom/Business Events)

Trigger on business events or custom events ingested into Grail.

Use Cases

- **Order completed** → Update inventory system
- **User registered** → Send welcome notification
- **Deployment finished** → Run validation tests

- **Custom alert** → External system integration

Configuration

```
trigger:
  type: event
  config:
    eventType: "com.company.order-completed"
    filterQuery: |
      event.type == "com.company.order-completed"
      AND order.total > 1000
```

Sending Business Events

Ingest events via API:

```
curl -X POST "https://&lt;env>/api/v2/bizevents/ingest" \
  -H "Authorization: Api-Token &lt;token>" \
  -H "Content-Type: application/json" \
  -d '{
    "type": "com.company.order-completed",
    "data": {
      "order_id": "ORD-12345",
      "customer": "ACME Corp",
      "total": 1500.00
    }
  }'
```

Event Data

Access event fields in tasks:

```
{{ event()["data"]["order_id"] }}
{{ event()["data"]["customer"] }}
{{ event()["data"]["total"] }}
```

7. Trigger Data and Expressions

Accessing Trigger Data

Expression	Returns	Example
<code>{{ event() }}</code>	Full event object	<code>{"title": "...", ...}</code>
<code>{{ event()["field"] }}</code>	Specific field	<code>"High response time"</code>
<code>{{ trigger() }}</code>	Trigger metadata	<code>{"type": "davis-problem"}</code>
<code>{{ trigger().params }}</code>	On-demand params	<code>{"env": "prod"}</code>

Common Event Fields by Trigger Type

Davis Problem:

```
{{ event()["display_id"] }}      # P-12345
{{ event()["title"] }}          # Problem title
{{ event()["severity"] }}       # CRITICAL, HIGH, MEDIUM, LOW
{{ event()["status"] }}        # OPEN, RESOLVED
{{ event()["affected_entity_ids"] }} # Array of entity IDs
{{ event()["root_cause_entity_id"] }} # Root cause entity
{{ event()["management_zones"] }} # Array of MZ names
{{ event()["problem_url"] }}    # Link to problem
```

Davis Event (Metric):

```
{{ event()["metric_key"] }}    # Metric identifier
{{ event()["value"] }}         # Current value
{{ event()["threshold"] }}     # Threshold that was breached
{{ event()["entity_id"] }}     # Affected entity
```

Schedule:

```
{{ trigger()["scheduled_time"] }} # When scheduled to run
{{ trigger()["actual_time"] }}    # When actually started
```

Query Davis Problems

```
// Recent Davis problems that could trigger workflows
fetch events, from: now() - 24h
| filter event.kind == "DAVIS_PROBLEM"
| fields timestamp,
          display_id,
          event.name,
          severity,
          status,
          affected_entity_ids,
          root_cause_entity_id
| sort timestamp desc
| limit 20
```

```
// Problem count by severity (last 7 days)
fetch events, from: now() - 7d
| filter event.kind == "DAVIS_PROBLEM"
| filter status == "OPEN"
| summarize problem_count = count(), by:{severity}
| sort problem_count desc
```

```
// Business events that could trigger workflows
fetch bizevents, from: now() - 24h
| summarize event_count = count(), by:{event.type}
| sort event_count desc
| limit 20
```

Next Steps

Now that you understand triggers, learn to send notifications:

Recommended Path

1. **WFLOW-03: Alert Notification Basics** - Slack, Teams, email notifications
2. **WFLOW-04: Advanced Notification Routing** - Conditional routing
3. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation patterns

Key Takeaways

- **Davis Problem** triggers for AI-detected incidents
 - **Davis Event** triggers for metric thresholds
 - **Schedule** triggers for recurring tasks
 - **On-Demand** triggers for testing and API integration
 - **Event** triggers for business events
 - Use expressions like `{{ event()["field"] }}` to access data
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Summary

In this notebook, you learned:

- All five trigger types and when to use each
 - How to configure Davis Problem triggers with filters
 - How to set metric thresholds with Davis Event triggers
 - Cron expressions for schedule triggers
 - On-demand execution via UI and API
 - Business event triggers for custom automation
 - How to access trigger data in expressions
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References

- [Workflow Triggers](#)
 - [Davis Problem Events](#)
 - [Business Events](#)
 - [Cron Expression Guide](#)
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official

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